4.19 Quiz: Normal and Binomial Distribution — Markscheme

1. A market stall sells bags of apples. It is known that 20% of the apples have bruises. A bag contains 10 randomly selected apples. Let X be the number of bruised apples in the bag.

   (a) Find the expected number of bruised apples in a bag, $E(X)$. [1]

   $E(X) = np = 10 \times 0.20 = 2$ **(A1)**

   (b) Find the probability that none of the apples are bruised. [2]

   $\mathrm{P}(X = 0) = \binom{10}{0}(0.20)^0(0.80)^{10}$ **(M1)**

   $= 0.80^{10} = 0.107$ **(A1)**

   (c) Find $\mathrm{P}(X = 3)$. [2]

   $\mathrm{P}(X = 3) = \binom{10}{3}(0.20)^3(0.80)^7$ **(M1)**

   $= 120 \times 0.008 \times 0.2097\ldots = 0.201$ **(A1)**

   GDC: binomPdf(10, 0.20, 3) $= 0.20133\ldots$

   (d) Find $\mathrm{P}(X \leq 3)$. [2]

   $\mathrm{P}(X \leq 3) = \mathrm{P}(X = 0) + \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3)$ **(M1)**

   $= 0.8791$ **(A1)**

   GDC: binomCdf(10, 0.20, 0, 3) $= 0.87913\ldots$

2. The marks on a mathematics test follow a normal distribution with mean $\mu = 72$ and standard deviation $\sigma = 8$.

   (a) Find the probability that a randomly selected student scores higher than 85. [2]

   $\mathrm{P}(M > 85)$ **(M1)**

   GDC: normCdf$(85,\ 10^{99},\ 72,\ 8) = 0.05208\ldots$

   ≈ 0.0521 **(A1)**

   (b) Find the probability that a student's mark is between 60 and 80. [2]

   $\mathrm{P}(60 < M < 80)$ **(M1)**

   GDC: normCdf$(60,\ 80,\ 72,\ 8) = 0.77453\ldots$

   ≈ 0.775 **(A1)**

   (c) 75% of the students score less than h marks. Find h. [3]

   $\mathrm{P}(M < h) = 0.75$ **(M1)**

   GDC: invNorm$(0.75,\ 72,\ 8)$ **(M1)**

   $h = 77.396\ldots \approx 77.4$ **(A1)**

3. The waiting time T (minutes) at a bus stop follows a normal distribution with mean 15 minutes and standard deviation σ minutes. It is known that 4% of waits are longer than 20 minutes.

   (a) Find the value of σ. [3]

   $\mathrm{P}(T > 20) = 0.04$, so $\mathrm{P}(T < 20) = 0.96$ **(M1)**

   GDC: invNorm$(0.96) = 1.7507\ldots$

   $\dfrac{20-15}{\sigma} = 1.7507$ **(M1)**

   $\sigma = \dfrac{5}{1.7507} = 2.856\ldots \approx 2.86$ **(A1)**

   (b) Find $\mathrm{P}(T < 12)$. [2]

   GDC: normCdf$(-10^{99}, 12, 15, 2.856)$ **(M1)**

   $= 0.1468\ldots \approx 0.147$ **(A1)**

   (c) On a particular day, 10 people wait at this bus stop. Find the probability that at most 2 of them wait more than 20 minutes. [3]

   $\mathrm{P}(T > 20) = 0.04$, so $Y \sim \mathrm{B}(10, 0.04)$ **(M1)**

   $\mathrm{P}(Y \leq 2)$ **(M1)**

   GDC: binomCdf$(10, 0.04, 0, 2) = 0.9938\ldots$

   ≈ 0.994 **(A1)**

   (d) A person is selected at random. Given that they waited less than 15 minutes, find the probability that they waited less than 10 minutes. [3]

   $\mathrm{P}(T < 10 \mid T < 15) = \dfrac{\mathrm{P}(T < 10)}{\mathrm{P}(T < 15)}$ **(M1)**

   $\mathrm{P}(T < 15) = 0.5$ (since $\mu = 15$) **(A1)**

   GDC: normCdf$(-10^{99}, 10, 15, 2.856) = 0.04006\ldots$

   $\mathrm{P}(T < 10 \mid T < 15) = \dfrac{0.04006}{0.5} = 0.0801\ldots \approx 0.0801$ **(A1)**

Total: 25 marks